Final Project: NYC Open Data Research Showcase

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0.1 Overview

For your final project, you will design and conduct an **original research project using NYC Open Data**.

Throughout this semester, you have learned how to import, clean, analyze, visualize, and communicate data using R and Quarto. Now it is time to bring those skills together!

You will select one of five approved NYC Open Data datasets, develop your own research question, and use statistical analysis to investigate it.

Your final project will be written as a fully reproducible **Quarto document**.

By the end of this project, you will have created a research report that you can publish online, share with others, and potentially include in your professional portfolio.

0.1.1 Learning Objectives

By completing this project, you will be able to:

- Develop an original, data-driven research question.
- Explore and analyze publicly available civic data.
- Clean and transform datasets using R.
- Select and apply appropriate statistical methods.
- Create meaningful data visualizations.
- Interpret statistical findings in plain English.
- Communicate research to a non-technical audience.
- Produce a fully reproducible Quarto document.
- Publish and share your research online.

0.2 Part 1: Choose Your Dataset

You must select **one of the five approved NYC Open Data datasets** below.

Each dataset explores a different aspect of life in New York City.

You may investigate **any research question you want**, provided that:

1. Your question can be answered using your selected dataset.
2. Your analysis uses statistical methods covered in this course.
3. Your research question is approved by your instructor.

The example questions below are intended to help you brainstorm. **You are not required to use any of them.**

> Working With NYC Open Data Using the nycOpenData Package
>
> Throughout this course, we have emphasized **reproducible research**, and your final project should follow the same principles.
> You are strongly encouraged to use the `nycOpenData` R package to access and explore your selected dataset directly in R.
> Using `nycOpenData` allows you to:
>
> - Access NYC Open Data datasets programmatically.
> - Explore available datasets and their variables.
> - Import data directly into your R workflow.
> - Create a reproducible analysis without manually downloading and editing CSV files.
>
> Don't forget to cite the package in your report using `@R-nycOpenData` and include the corresponding entry in your `packages.bib` file!

0.2.1 Option 1: NYC 311 Service Requests

Dataset: 311 Service Requests from 2020 to Present

New Yorkers use NYC 311 to report problems and request non-emergency city services.

The dataset includes information about complaints, government agencies, locations, and dates.

Example research questions:

- Do certain boroughs report different types of complaints?

- Are noise complaints more common during particular months?
- Do complaint patterns change throughout the year?
- Are certain types of service requests resolved more quickly than others?
- How do complaints differ across NYC neighborhoods?

Possible statistical methods:

- Chi-square tests
- Descriptive statistics
- Correlation
- Linear regression
- Grouped comparisons

0.2.2 Option 2: NYC Restaurant Inspections

Dataset: DOHMH New York City Restaurant Inspection Results

The NYC Department of Health and Mental Hygiene conducts restaurant inspections to evaluate compliance with health regulations.

This dataset includes restaurant information, inspection results, violations, and scores.

Example research questions:

- Do restaurant inspection scores differ across boroughs?
- Are certain violations more common in particular cuisines?
- Is there a relationship between inspection scores and inspection outcomes?
- Do inspection scores change over time?
- Are certain violation categories more common than others?

Possible statistical methods:

- Chi-square tests
- T-tests
- ANOVA
- Correlation
- Linear regression

Important: Restaurants may appear multiple times because they can have multiple inspections and violations. Consider this when determining what each row represents and how to analyze the data.

0.2.3 Option 3: NYC Payroll

Dataset: Citywide Payroll Data (Fiscal Year)

New York City employs people across many different government agencies and occupations.

This dataset contains information about city employees, agencies, job titles, and compensation.

Example research questions:

- Do average salaries differ across city agencies?
- Is there a relationship between regular gross pay and overtime pay?
- Do certain job titles receive more overtime compensation?
- How have salaries changed across fiscal years?
- How does compensation vary across occupational categories?

Possible statistical methods:

- Descriptive statistics
- T-tests
- ANOVA
- Correlation
- Linear regression

Important: Pay fields may represent different types of compensation, and the data may contain records from multiple fiscal years. Define your variables and population carefully.

0.2.4 Option 4: NYC Motor Vehicle Crashes

Dataset: Motor Vehicle Collisions – Crashes

This dataset contains information about reported motor vehicle collisions throughout New York City.

It includes crash dates, locations, injuries, fatalities, and contributing factors.

Example research questions:

- Do crash frequencies differ across boroughs?
- Are crashes more common during certain times of day?
- Is there a relationship between crash timing and the number of injuries?
- Do contributing factors differ across boroughs?
- Are crashes involving injuries more common during particular days of the week?

Possible statistical methods:

- Chi-square tests
- T-tests
- ANOVA
- Correlation
- Linear regression

Important: This dataset describes reported crashes, not all driving activity. A higher number of crashes does not automatically mean a location is more dangerous without considering traffic exposure.

0.2.5 Option 5: NYC Parking and Camera Violations

Dataset: Open Parking and Camera Violations

New York City issues parking and camera-based traffic violations.

This dataset contains records of open violations, including information about fines, violation types, and other available characteristics.

Example research questions:

- Which types of violations have the highest average fines?
- Do fine amounts differ across violation categories?

- Are certain violations more common than others?
- Is there a relationship between original fines and outstanding amounts?
- How do violation patterns differ across available geographic categories?

Possible statistical methods:

- Chi-square tests
- T-tests
- ANOVA
- Correlation
- Linear regression

Important: This dataset contains open violations. It should not automatically be interpreted as representing every violation issued in NYC. Review the available fields before developing your research question.

0.3 Part 2: Develop Your Research Question

Once you have selected your dataset, you must develop an **original research question**.

Your research question should be:

- **Specific:** Clearly identify what you want to investigate.
- **Measurable:** Answerable using variables in your dataset.
- **Statistical:** Suitable for at least one statistical method covered in class.
- **Meaningful:** Connected to something interesting or important about NYC.

For example:

Too broad:

> Are NYC restaurants safe?

More specific:

> Do average restaurant inspection scores differ across NYC boroughs?

The second question identifies measurable variables and can be investigated using statistical analysis.

You are encouraged to be creative!

However, your question must be realistic given the available data and the methods you have learned.

0.4 Part 3: Research Proposal Checkpoint

Before beginning your full analysis, you must submit a brief research proposal for instructor approval.

Your proposal should include:

0.4.1 1. Selected Dataset

Identify which of the five approved datasets you selected.

Include a hyperlink to the dataset.

0.4.2 2. Research Question

Write your research question in one or two sentences.

0.4.3 3. Variables

Identify the variables you plan to investigate.

Explain what each variable represents.

0.4.4 4. Statistical Method

Identify at least one statistical method you plan to use.

Examples include:

- Chi-square test
- Independent-samples t-test
- ANOVA
- Correlation
- Linear regression

Explain briefly why the method is appropriate.

0.4.5 5. Planned Visualization

Describe at least one visualization you plan to create.

Examples include a bar chart, scatterplot, boxplot, or line chart.

Explain what you hope the visualization will communicate.

Your instructor must approve your proposal before you proceed with the full project.

You may revise your research question or analysis plan after approval, but significant changes should be discussed with your instructor.

0.5 Part 4: Final Quarto Research Report

Your final research report must be written using **Quarto and R**.

Your report should combine written explanations, R code, statistical results, and visualizations into one cohesive document.

The goal is not simply to generate statistical output.

Your goal is to use data to answer a question and communicate what you discovered.

Your document must contain the following sections.

0.5.1 1. Introduction

Introduce your topic and research question.

Your introduction should explain:

- What you are investigating.
- Why you selected this topic.
- Why the question is interesting or important.
- What you hope to learn.

Clearly state your research question.

Write for a reader who may not have a background in statistics.

0.5.2 2. Dataset Description

Introduce your selected dataset.

Include:

- The dataset name.
- A hyperlink to its NYC Open Data page.
- A brief explanation of what the dataset contains.
- The variables relevant to your research question.
- The time period covered by your analysis.
- The number of observations included in your analysis.

Explain why the dataset is appropriate for answering your research question.

0.5.3 3. Data Import and Preparation

Import your data using R.

Demonstrate appropriate data preparation techniques covered in class.

Depending on your dataset, this may include:

- Selecting relevant variables.
- Filtering observations.
- Handling missing values.
- Creating new variables.
- Grouping observations.
- Calculating summary statistics.
- Converting variables into appropriate data types.

Explain the important cleaning and transformation decisions you make.

You do not need to perform every type of cleaning listed above.

Only perform transformations that are relevant to your research question.

Your final workflow must be reproducible. Avoid manually editing your dataset outside R.

0.5.3.1 Working With Large Datasets

Some approved NYC Open Data datasets contain millions of observations.

You are **not required to analyze the entire dataset**.

You may work with a smaller, clearly defined subset of the data.

For example, you might investigate:

- One calendar year.
- One borough.
- A particular category of complaints.
- A specific group of government agencies.
- A defined period of restaurant inspections.

If you use a subset, explain how you selected it and why.

Your data import and filtering process must be documented so another person can reproduce your analysis.

0.5.4 4. Descriptive Analysis

Before conducting statistical tests, explore your data.

Include **at least one descriptive analysis** relevant to your research question.

Examples include:

- Frequencies and percentages.
- Means and medians.
- Standard deviations.
- Grouped summary statistics.
- Minimum and maximum values.

Present relevant results using a properly formatted table or visualization.

Explain what the descriptive results reveal about your data.

0.5.5 5. Statistical Analysis

You must conduct **at least one statistical analysis** using a method covered in this course.

Approved methods include:

- Chi-square test
- Independent-samples t-test
- ANOVA
- Pearson correlation
- Simple linear regression

Select a method that is appropriate for your research question and variables.

Your analysis must include:

1. **Research question:** What are you investigating?
2. **Method:** Which statistical method did you select?

3. **Justification:** Why is this method appropriate?
4. **Results:** What did the statistical analysis reveal?
5. **Interpretation:** What do these results mean in plain English?

Where appropriate, report relevant statistical information such as:

- Test statistics
- P-values
- Means or group differences
- Correlation coefficients
- Regression coefficients
- R-squared

You should also consider the assumptions and limitations of your chosen method.

Important: Statistical significance does not necessarily mean that a result is practically important, and correlation does not establish causation.

Your interpretation should reflect what the data can reasonably support.

> **!** Statistical Analysis Requirement
>
> Descriptive statistics and visualizations alone do not satisfy the statistical analysis requirement.
> You must apply at least one of the approved statistical methods covered in class and explain why it is appropriate for your research question and data.

0.5.6 6. Data Visualizations

Your report must include **at least two distinct visualizations created using `ggplot2`**.

Your visualizations should help answer your research question.

Examples include:

- Bar charts
- Histograms
- Boxplots
- Scatterplots

- Line charts

Each visualization must:

- Have a meaningful title.
- Include appropriate axis labels.
- Use readable text and formatting.
- Include a caption.
- Be relevant to your research question.
- Be discussed in the written narrative.

Your visualizations should communicate different information rather than repeating the same result.

For example, one visualization might show the overall distribution of a variable, while another compares that variable across groups.

0.5.7 7. Discussion and Interpretation

This is one of the most important parts of your project.

Explain what you learned from your analysis.

Your discussion should address:

- What were your main findings?
- Did the results answer your research question?
- Were any results surprising?
- What limitations should readers consider?
- What additional questions could be investigated?

Avoid simply repeating numerical results.

Instead, explain what those results mean.

For example:

Weak interpretation:

> The p-value was 0.02.

Stronger interpretation:

> The statistical test suggests that restaurant inspection scores differ across the groups examined. However, the results do not establish why these differences exist.

Your interpretation should be understandable to someone unfamiliar with statistical terminology.

0.5.8 8. Civic Relevance and Open Data

Explain why your findings matter beyond the classroom.

Consider:

- Who might be interested in your research?
- What could New Yorkers learn from your findings?
- Could the results be useful to journalists, researchers, community organizations, or policymakers?
- How does publicly available data make this research possible?

Connect your project to the broader value of open data.

0.5.9 9. Conclusion

End your report with a brief conclusion.

Summarize:

- Your original research question.
- Your most important finding.
- What your analysis contributes to understanding NYC.
- One possible direction for future research.

Your conclusion should leave readers with a clear understanding of what you discovered.

0.6 Part 5: Quarto and Reproducibility Requirements

Your final report must demonstrate the reproducible research skills developed throughout the semester.

0.6.1 Quarto Document

Your report must:

- Be written in a `.qmd` file.
- Include narrative, code, and output in one document.
- Render successfully to HTML.
- Suppress unnecessary warnings and messages.
- Use meaningful section headings.
- Include clearly labeled code chunks.

0.6.2 Figures and Tables

All figures and tables must be properly labeled and referenced.

Use Quarto labels beginning with:

- `fig-` for figures.
- `tbl-` for tables.

For example:

`fig-restaurant-scores`

`tbl-summary-statistics`

You must refer to figures and tables within your written narrative using Quarto cross-references.

For example:

> As shown in @fig-restaurant-scores, inspection scores vary across the groups examined.

Or:

> @tbl-summary-statistics presents the descriptive statistics for the variables included in the analysis.

Every figure and table must be discussed in your narrative.

Tables should be formatted using `knitr::kable()` or another appropriate method demonstrated in class.

0.6.3 Citations and References

Your report must properly cite:

- Your selected NYC Open Data dataset.
- R packages used in your analysis.
- Any additional external sources used.

Include a bibliography file named `packages.bib`.

Use package citations such as:

`@R-ggplot2`

`@R-dplyr`

`@R-knitr`

Your bibliography should include the corresponding entries for packages you cite.

All external information must be appropriately attributed.

0.6.4 Reproducible Code

Your document must:

- Load required packages explicitly.
- Import data using R code.
- Document important cleaning decisions.
- Avoid manual modifications to the source data.
- Use relative file paths where applicable.
- Avoid hard-coding statistical results.
- Set a random seed when using randomness.
- Render without errors.

A reader should be able to follow your workflow from data import to final results.

0.7 Part 6: Length and Formatting

Your report must include:

- **350–500 words of written narrative**, excluding code, tables, captions, and references.
- At least one descriptive analysis.
- At least one appropriate statistical analysis.
- At least two `ggplot2` visualizations.
- Properly labeled and cross-referenced figures and tables.
- A discussion of findings and limitations.
- Proper citations and a bibliography.
- A cleanly rendered HTML document.

The word count is a target range. Clear, substantive explanations are more important than filling space.

You are welcome to exceed 500 words if additional explanation improves your report.

0.8 Part 7: Publish Your Research

Once your report is complete, publish your HTML document using **Posit Connect Cloud**.

Your published report should:

- Be accessible through a working public link.
- Display all figures and tables correctly.
- Include the complete research narrative.
- Render without visible errors.

You will submit the published link along with your project files.

The goal is to create something that can be shared beyond the classroom.

0.9 Part 8: Class Publication Opportunity

At the end of the semester, selected or approved student projects may be compiled into a **collaborative Quarto book**.

Each participating student's research project could become its own chapter.

The final book could showcase how students used public data to investigate different aspects of life in New York City.

This would create:

- A public collection of student research.
- A portfolio artifact students can share.
- An opportunity to demonstrate reproducible research.
- A collaborative educational resource.

Participation in public publication is optional.

Students may opt out without any impact on their grade.

Outstanding projects may also provide a foundation for future presentations or submissions to **NYC Open Data Week 2027**.

0.10 Grading Rubric

Your final project will be graded out of **100 points**.

Category	Points	Expectations
Research Question and Civic Relevance	15	Clear, original, appropriate question; completed proposal checkpoint; meaningful civic connection
Data Preparation and Reproducibility	20	Correct data import, appropriate cleaning, transparent workflow, reproducible code

Category	Points	Expectations
Statistical Analysis and Interpretation	30	Appropriate statistical method, correct execution, clear justification, meaningful interpretation, consideration of assumptions and limitations
Visualizations and Cross-References	15	Two informative visualizations, clear labels, captions, correct Quarto cross-references
Writing, Citations, and Quarto Formatting	20	Organized narrative, appropriate citations, bibliography, clean HTML output, professional presentation of written research
Total	**100**	

Remember: You are being evaluated on the quality of your research process, analysis, and interpretation—not whether your results are statistically significant.

0.11 Submission Checklist

Before submitting, make sure you have completed everything below.

0.11.1 Research

- ☐ Selected one of the five approved datasets.
- ☐ Developed an original research question.
- ☐ Submitted and received approval for the research proposal.
- ☐ Identified the relevant variables.
- ☐ Completed at least one descriptive analysis.
- ☐ Completed at least one appropriate statistical analysis.

☐ Interpreted the results in plain English.

☐ Discussed limitations and civic relevance.

0.11.2 Quarto

☐ Included all required report sections.

☐ Created at least two `ggplot2` visualizations.

☐ Added appropriate figure and table labels.

☐ Cross-referenced all figures and tables.

☐ Formatted tables appropriately.

☐ Included package citations.

☐ Included `packages.bib`.

☐ Documented data preparation decisions.

☐ Confirmed the document renders without errors.

0.11.3 Publication

☐ Published the report to Posit Connect Cloud.

☐ Tested the published link.

0.12 Final Submission

Submit the following:

1. Your completed `.qmd` file.
2. Your `packages.bib` file.
3. Your public Posit Connect Cloud link.
4. Any local data files necessary to reproduce your analysis, unless your document imports the data directly from a stable public source.

0.13 Resources

NYC Open Data

- NYC Open Data Portal
- NYC 311 Service Requests
- NYC Restaurant Inspections
- NYC Payroll
- NYC Motor Vehicle Crashes
- NYC Parking and Camera Violations

Course Materials

- Reproducible Research Using R
- Quarto Documentation
- Posit Connect Cloud

0.14 Final Takeaway

Throughout this course, you have learned how to work with data, apply statistical methods, create visualizations, and communicate results.

Now you have the opportunity to bring those skills together in a project that is entirely your own.

There is no single correct research question, and there is no requirement that your analysis produce a statistically significant result.

What matters is that you ask a meaningful question, use appropriate methods, and communicate what the evidence reveals.

Your final project is more than an assignment. It is an opportunity to contribute your own perspective to understanding New York City through data.